# Bridge Day 2 INSTRUCTOR KEY

Variables, Arithmetic, Conversion, and Formatted Output

Tuesday, July 7, 2026 • 20 points • target 18 minutes

| Name | | UIN | | Section | |
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## Assessment boundary and evidence source

**Contract:** 07a04826c975
**Cumulative foundations:** print, assignment, integer arithmetic, input conversion, f-strings
**Scaffold level:** complete two-input calculation program
**Evidence source:** the matching v5 workbook readiness/evidence pages and VIP activities 18.3, 18.4, 18.5, 18.6, 18.7. Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

## Item 1. Trace quotient, remainder, and final state. - 4 points

```
minutes = 367
hours = minutes // 60
remaining = minutes % 60
code = hours * 10 + remaining
print(hours, remaining, code)
```

**Question:** Show each arithmetic result and give the exact output. **Model response:** 367 // 60 is 6 and 367 % 60 is 7. The final expression stores 6 * 10 + 7 = 67. Exact output: 6 7 67.

## Item 2. Distinguish text combination from numeric addition. - 4 points

```
raw = "8"
extra = 3
print(raw + str(extra))
print(int(raw) + extra)
```

**Question:** Name the type used by each + operation and give both exact output lines. **Model response:** The first + combines two strings, so it produces 83. The second converts raw to integer 8 and performs numeric addition, producing 11. Exact output lines: 83 and 11.

## Item 3. Repair a missing input conversion. - 4 points

```
kits = "4"
total = kits * 6
print(f"Total items:  {total}")
```

**Question:** Give the actual output, identify the stored type that caused it, and write the minimal repair. **Model response:** kits is a string, so multiplying by 6 repeats the character six times. Actual output is Total items: 444444. Repair with kits = int(kits) before the calculation, which produces Total items: 24.

# Complete program - 8 points

- Read length and width as floating-point inputs.
- Compute rectangle area and perimeter.
- Print Area: followed by one decimal place, then Perimeter: followed by one decimal place.
- The program must work for values other than the example.

### Model program

```
length = float(input())
width = float(input())

area = length * width
perimeter = 2 * (length + width)

print(f"Area:  {area:.1f}")
print(f"Perimeter:  {perimeter:.1f}")
```

### Reasoning and test evidence

- Both input results are text until float performs conversion. The calculations use the converted variables.
- For inputs 3 and 5, area is 15.0 and perimeter is 16.0, so the required output lines are Area: 15.0 and Perimeter: 16.0.

**Rubric:** 2 input/conversion, 2 named calculations, 2 correct formulas, 2 exact f-string labels and precision.

Quiz evidence is sampled from the taught construct boundary; workbook and notebook evidence remain the primary learning surfaces.